

TBA4245 Geodesy

Example exercise no. 6. Ionosphere

Some pages on www especially about the ionosphere, see:

http://iri.gsfc.nasa.gov Check the “Related links”, also shown in the powerpoint presentation in the lectures.
https://nssdc.gsfc.nasa.gov/space https://ccmc.gsfc.nasa.gov/modelweb/models/iri2016_vitmo.php Ionospheric model which can be run on www The International Reference Ionosphere (IRI) is an international project.
http://iono.jpl.nasa.gov/gim.html JPL-NASA, central laboratory for GPS, this address gives some of the figures used in the lectures
http://sidc.oma.be/ Royal Observatory of Belgium harbours the Sunspot Index Data centre, the World data centre for the Sunspot Index.

(a) Ionosphere models

- (a.1) Describe in few words IRI (International Reference Ionosphere), which you will find here:

<http://iri.gsfc.nasa.gov/>

https://ccmc.gsfc.nasa.gov/modelweb/models/iri2016_vitmo.php

go to

International Reference Ionosphere, web version from 2016:

https://ccmc.gsfc.nasa.gov/modelweb/models/iri2016_vitmo.php

The program computes several parameters valid for the ionosphere. We will only look into TEC (total electron content). Use 2000 km as the height in this model. NB also in ”optional input” to obtain the numerical values of TEC. Make some printouts showing some results from the program.

See the next page for more info of how to use the program.

- (a.2) Make a 24 hours TEC profile for Trondheim for today (”hour profile”).
- (a.3) Make a TEC profile for Trondheim covering several years, from 1980 up to today (”year profile”). Use the time of the day when the maximum TEC occurs.
- (a.4) Make a ”latitude” TEC profile for the meridian through Trondheim (from pole to pole) to see if there are any differences in TEC depending on the latitude. Use the time of the day when the maximum TEC occurs.
- (a.5) Compare your results with a regional or global TEC map. See the the table on top of this page for actual web pages. Or google!

Choices made in the 5 boxes to answer question (a.2).

For (a.3) and (a.4) you must change some parameters in box 1 (the date and coordinates (if necessary) and in last line (profile type and parameters)) and in box 4 (output parameters)

Select Date and Time
Year(1958-2016): 2002
Note:If date is outside the Ap index range (1958-2013/10), then STORM model will be turned off.
Month: January Day(1-31): 01
Time: Universal Hour of day (e.g. 1.5): 12.
Select Coordinates
Coordinates Type: Geographic
Latitude(deg.,from -90. to 90.): 63. Longitude(deg.,from 0. to 360.): 10.
Height (km, from 60. to 2000.): 2000.
Select a Profile type and its parameters:
Hour profile[0-24] Start 0 Stop 24 Stepsize 1.

Optional Input:
Sunspot number, Rz12 (0. - 400.) Ionospheric index, IG12 (-50. - 400.)
F10.7 radio flux, daily (0. - 400.) F10.7 radio flux, 81-day (0. - 400.)
Electron content: Upper boundary (km., from 50. - 2000.) 2000.
Ne Topside: NeQuick F peak model: URSI foF2 Storm model: on

Select output form:
☐ List model data
☐ Create model data file in ASCII format for downloading
☒ Plot model data

Select desired output parameters
Independent Variables
☐ Year
☐ Month
☐ Day of month
☐ Day of year
☒ Hour of day, UT/LT
(depending on user's choice above)
☐ CGM Latitude, deg.
☐ CGM Longitude, deg.
☐ Magnetic inclination (DIP), degree
☐ Modified dip latitude, degree
☐ Declination, degree

IRI Model Parameters
☐ Electron_density (Ne), m^{-3}
☐ Ratio of Ne and F2 peak density(Ne/NmF2)>
☐ Neutral Temperature Tn, K
☐ Ion Temperature Ti, K
☐ Electron Temperature, Te, K
☐ Atomic Oxygen ions (O^+), percentage
☐ Atomic Hydrogen (H^+), ions, percentage
☐ Atomic Helium (He^+), ions, percentage
☐ Molecular Oxygen (O_2^+) ions, percentage
☐ Nitric Oxide ions (NO^+), percentage
☐ Cluster ions, percentage
☒ Atomic Nitrogen (N^+) ions, percentage
☒ Total Electron Content (TEC), $10^{16} m^{-2}$
☐ TEC top, percentage

(b) Ionospheric delay calculations

Assume two cases of ionospheric noise:

- "Normal" or small value of TEC: $2 \cdot 10^{16} \text{ el/m}^2$
- Very high value of TEC: $100 \cdot 10^{16} \text{ el/m}^2$

L1=1575,42 MHz

L2=1227,60 MHz

c=299792458 m/s

Hint: See the GNSS book, section 5.3.2, especially pages 119 and 120, for formulas.

Useful formula: $\lambda = c / f$

(b1) Phase measurements:

Calculate the ionospheric delays in meter (with sign!) for both the carrier waves L1 and L2.

Use the two TEC values given above.

The TEC values are varying from time to time. They changes arbitrary (random) and are therefore difficult to model. Assume that there is a change in the TEC value of 1 % over a short period and that this change in value is not detected by and corrected in the modelling.

Calculate how much the ionospheric delay is changing because of the 1% change in the TEC value.

2 TECU: 1 % change

100 TECU: 1 % change is equal to 1 TECU

Is this change so large that it will cause problems for the baseline computations using the phase measurements? Hint: Compare the size of the change with what?

Phase measurements, fill in table below:

	Wave length	Delays		1 % change	
Carrier wave	λ	Normal TEC	High TEC	Normal TEC	High TEC (1 % = 1 TECU)
L1					
L2					

(b2) Pseudo range measurements (code measurements):

Calculate the ionospheric delays in meter (with sign!).

Use the two TEC values given above.

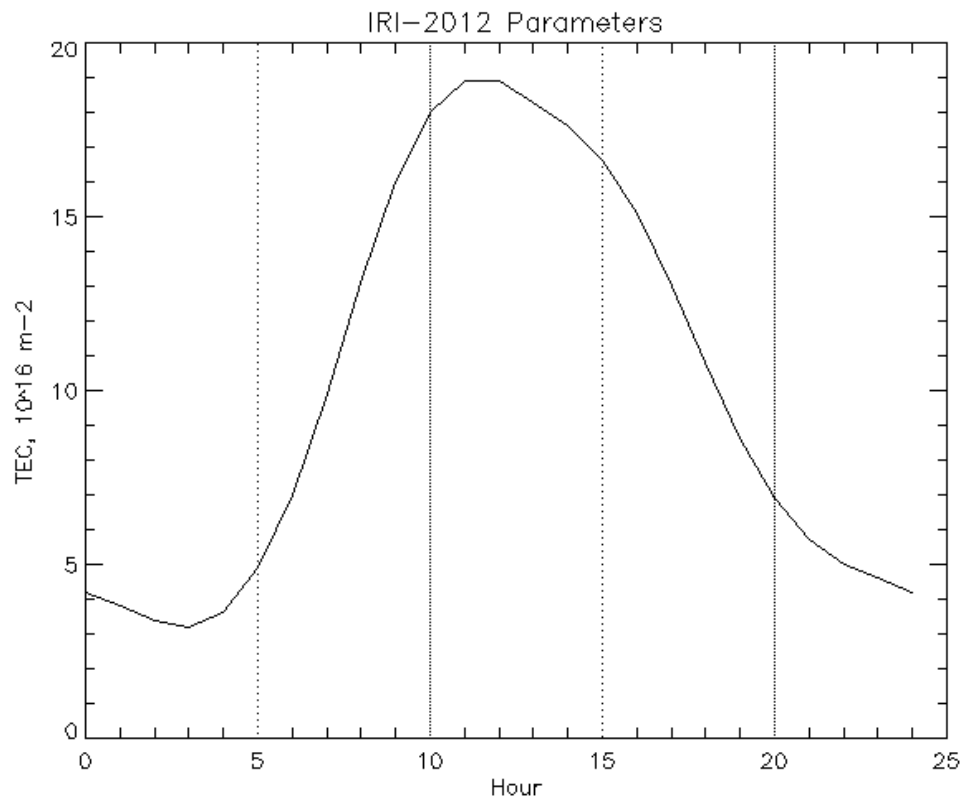
Calculate how much the ionospheric delay is changing because of the 1% change in the TEC value.

SOLUTION

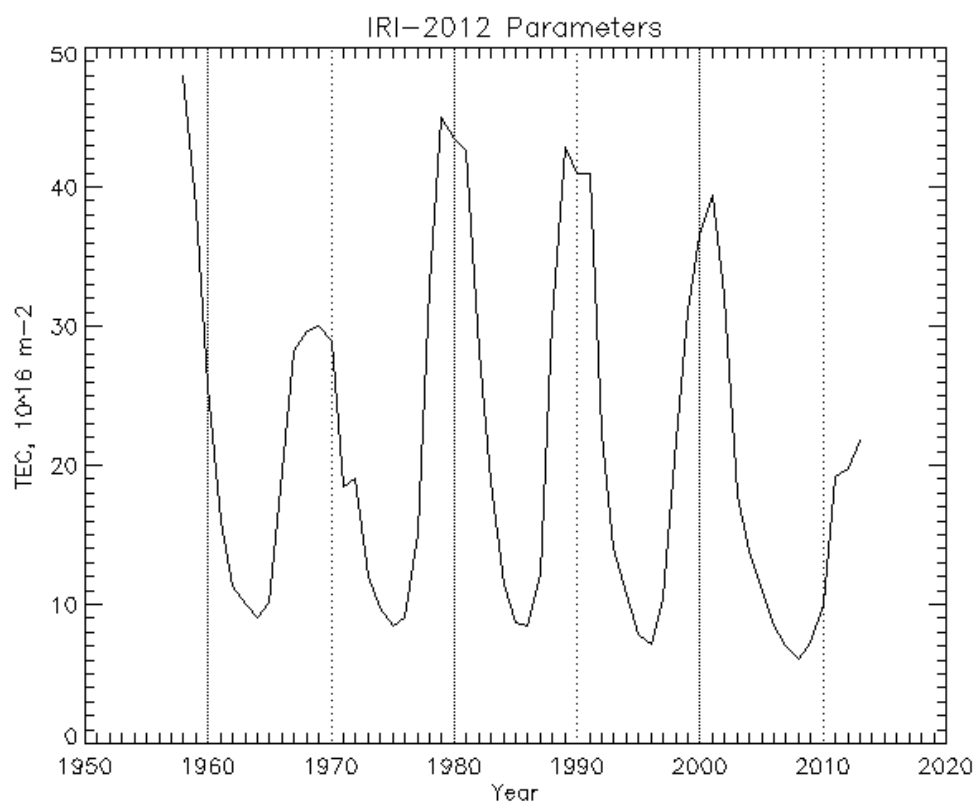
Question (a):

(a.1) See textbook or [www](#).

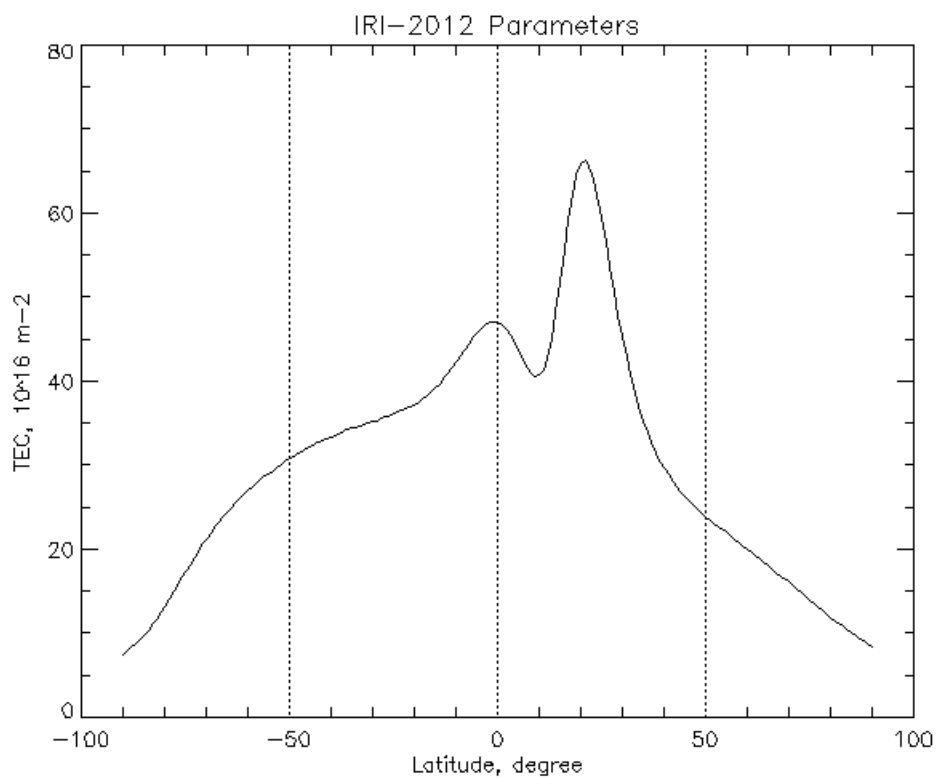
(a.2) TEC-plot. 30.9.2013, 0 -24 hours



(a.3) TEC, cycles 1958-2013 (30.9.2013)

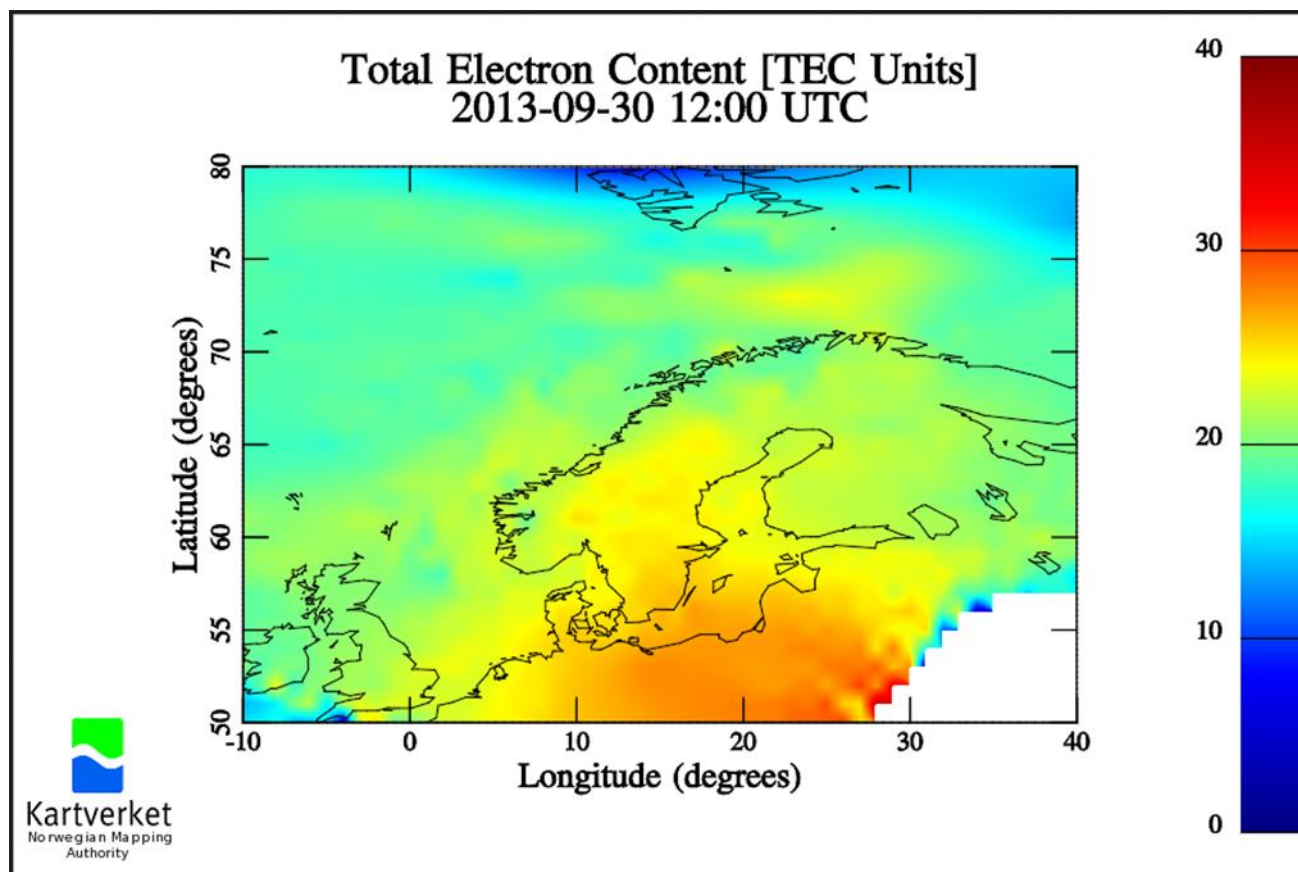


(a.4) TEC-plot 30.9.2013, latitude (along a meridian) Longitude = 10°



(a.5) TEC map:

<http://sesolstorm.kartverket.no/> 30.9.2013, 12 UTC



Question (b):

Calculations of delays through the ionosphere:

"Normal" value of TEC: $2 \cdot 10^{16}$ electrons/m²

Very high value of TEC: $100 \cdot 10^{16} = 100 \text{ TECU el/m}^2$

(b.1) Phase measurements:

	Wave length	Delays		1 % change	
Carrier wave	λ	Normal TEC	High TEC	Normal TEC	High TEC (1 % = 1 TECU)
L1	0,190 m	-0,325 m	-16,237 m	0,003 m	0,162 m
L2	0,242 m	-0,535 m	-26,742 m	0,005 m	0,267 m

Frequency L1: 1575,42 MHz

Frequency L2: 1227,60 MHz

$$\text{L1: } \Delta_{ph}^{iono} = -\frac{40,3 \cdot 100 \cdot 10^{16}}{(1,57542 \cdot 10^9)^2} = 16,237 \text{ m}$$

$$\Delta_{ph}^{iono} = -\frac{40,3}{f^2} \cdot TEC$$

Problems for the phase measurements and the baseline processing?

We calculate a small value for the normal TEC condition (3 and 5 mm) which is not significant, but a large value for high TEC condition (16.2 and 26.7 cm).

The effect of 1% change in TEC value when we have a high TEC value of 100 TECU is ca. the same as one period/cycle of the carrier waves L1 and L2. This is significant in GNSS positioning and this error will directly affect the measured distance between satellite and receiver and then the baseline computations.

(b.2) Code measurements:

Same numerical value, but opposite sign, +

$$\Delta_{gr}^{iono} = +\frac{40,3}{f^2} \cdot TEC$$